THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET C

Chapters 13 & 14: **Two Variables, One Line, Math of Space: Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The graph of the equation x = 5 is a line that is: 1

(a) parallel to the y-axis **(b)** having slope 5

(c) parallel to the x-axis **(d)** passing through the origin

2. A line with a positive slope: 1

(a) is vertical **(b)** rises as we move from left to right

(c) falls as we move from left to right **(d)** is horizontal

3. **Assertion (A):** The pair x + y = 4 and 2x + 2y = 8 has infinitely many solutions. 1

Reason (R): Every pair of linear equations in two variables has either exactly one solution or infinitely many solutions.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

4. The solution of the pair 3x + y = 11 and 2x − y = -1 is: 1

(a) (-2, 5) **(b)** (2, 5) **(c)** (5, 2) **(d)** (2, -5)

5. The slope of the line through the points (1, 2) and (3, 8) is: 1

(a) 4 **(b)** 1/3 **(c)** -3 **(d)** 3

6. The total surface area of a cuboid of length 9 cm, width 7 cm and height 2 cm is: 1

(a) 105 cm^2 **(b)** 95 cm^2 **(c)** 126 cm^2 **(d)** 190 cm^2

7. The volume of a cube is 27 cm^3. Its total surface area is: 1

(a) 54 cm^2 **(b)** 27 cm^2 **(c)** 36 cm^2 **(d)** 9 cm^2

8. Taking $\pi = \frac{22}{7}$, the curved surface area of a cylinder of radius 3.5 cm and height 12 cm is: 1

(a) 286 cm^2 **(b)** 462 cm^2 **(c)** 264 cm^2 **(d)** 132 cm^2

9. Taking $\pi = \frac{22}{7}$, the volume of a cylinder of radius 3.5 cm and height 8 cm is: 1

(a) 154 cm^3 **(b)** 176 cm^3 **(c)** 616 cm^3 **(d)** 308 cm^3

10. Taking $\pi = \frac{22}{7}$, the volume of a cone of base radius 6 cm and height 7 cm is: 1

(a) 264 cm^3 **(b)** 286 cm^3 **(c)** 792 cm^3 **(d)** 132 cm^3

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Show that $x + 2y = 4$ and $3x + 6y = 12$ have the same solutions. 2

12. The point $(1, k)$ lies on $3x + 2y = 7$. Find k. 2

13. What is a cross-section of a solid? What shape is the cross-section of a cylinder when it is cut parallel to its base? 2

14. Find the volume of a pyramid with a square base of side 5 cm and a height of 12 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Line l_1 passes through $(0, 2)$ and $(3, 8)$; line l_2 passes through $(0, -4)$ and $(2, 0)$. Find the slope of each line and say whether the pair of equations has a solution. 3

16. The areas of three faces of a cuboid that meet at a corner are 12 cm^2, 20 cm^2 and 15 cm^2. Find its volume. 3

17. Show that the lines $y = mx + d_1$ and $y = mx + d_2$ with $d_1 \neq d_2$ are parallel. 3

OR

2 adult and 3 child tickets cost ₹1050; 3 adult and 2 child tickets cost ₹1200. Find the cost of one adult ticket and one child ticket. 3

18. A steel tumbler is open at the top and has radius 3.5 cm and height 10 cm. Find the area of steel used in making it ($\pi = \frac{22}{7}$). 3

OR

A cylinder and a cone have the same base radius 7 cm and the same height 12 cm. Find the volume of each and the difference between them ($\pi = \frac{22}{7}$). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. State the three possible cases of the pair $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ in terms of the ratios $\frac{a_1}{a_2}$, $\frac{b_1}{b_2}$, $\frac{c_1}{c_2}$ and the graphs. Apply them to $4x - 2y = 6$ with $6x - 3y = 10$. 5

OR

Prove that the slope of $y = mx + d$ is m using any two points on the line, and use it to find the slope and y-intercept of $6x - 3y + 9 = 0$. 5

20. **Case study:** A swimming pool is a cuboid 20 m long, 10 m wide and 1.8 m deep. 5

(i) Find its capacity in litres.

(ii) Find the area of the floor and four walls to be tiled.

(iii) Find the tiling cost at ₹250 per m^2.

(iv) A pump fills 3000 litres per minute; find the time to fill the pool.

OR

A cylindrical jar of radius 7 cm contains water. A solid sphere of radius 3.5 cm is dropped in and sinks completely without any water overflowing ($\pi = \frac{22}{7}$). 5

(i) Find the volume of the sphere.

(ii) By how much does the water level rise?

(iii) By how much would it rise if three such spheres were dropped in?

(iv) Explain why the rise depends on the jar's base area.

END OF QUESTION PAPER — **Total: 40 Marks**